Calculator Model / No.

Name ________________________ **Class:** ___________ **Reg Number:** _____

MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 1

Chemistry 8872/01

Paper 1 Multiple Choice **20 September 2012**

50 minutes

Additional Materials: *OMR Sheet*
Data Booklet

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.
Write your calculator brand and model/number in the box provided above.

There are **thirty** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the OMR answer sheet.

Read very carefully the instructions on the use of the OMR answer sheet.

You are advised to fill in the OMR Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Use of OMR Answer Sheet

Ensure you have written your name, class register number and class on the OMR Answer Sheet.

Use a **2B** pencil to shade your answers on the OMR sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the ***OMR sheet***, please follow the given examples:
If your register number is **1**, then shade **01** in the index number column.
If your register number is **21**, then shade **21** in the index number column.

This document consists of **13** printed pages.

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Section A

For each question there are for possible answers, **A**, **B**, **C** and **D**. Choose the one you consider to be correct.

1 Which of the following statements contains one mole of the stated particle?

A atoms in 38.0 g of fluorine gas

B electrons in 24.0 dm^3 of hydrogen gas at room temperature and pressure

C neutrons in 1.00 g of helium gas

D protons in 2.02 g of neon gas

2 *Use of the Data Booklet is relevant to this question.*

A typical solid fertiliser for use with household plants and shrubs contains the elements N, P, and K in ratio of 15 g : 30 g: 15 g per 100 g of fertiliser. The recommended usage of fertiliser is 14 g of fertiliser per 5 dm^{-3} of water.

What is the concentration of nitrogen atoms in this recommended solution?

A 0.03 mol dm^{-3}

B 0.05 mol dm^{-3}

C 0.42 mol dm^{-3}

D 0.75 mol dm^{-3}

3 A barium salt, $BaFeO_n$, can react with hydrochloric acid to produce iron(III) ions and chlorine gas. In an experiment, upon treatment with excess hydrochloric acid, two moles of $BaFeO_n$ reacted to yield three moles of chlorine.

What is the value of $\boldsymbol{n}$?

A 1

B 2

C 4

D 6

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4 Two elements **X** and **Y** have the following properties.

- **X** and **Y** form ionic compounds Na_2**X** and Na_2**Y** respectively.
- Element **Y** forms **Y**F_6 molecule whereas **X** is unable to do so.

Which pair of electronic configurations of **X** and **Y** is correct?

	X	**Y**
A	[He] $2s^2$ $2p^2$	[Ne] $3s^2$ $3p^2$
B	[Ne] $3s^2$ $3p^2$	[He] $2s^2$ $2p^2$
C	[He] $2s^2$ $2p^4$	[Ne] $3s^2$ $3p^4$
D	[Ne] $3s^2$ $3p^4$	[He] $2s^2$ $2p^4$

5 Ions of the two most common isotopes of zinc are shown below:

$^{64}_{30}Zn^{2+}$ $^{66}_{30}Zn^{2+}$

Which of the following statements is correct?

A Both these Zn^{2+} ions have the same number of electrons but different number of protons.

B Both these Zn^{2+} ions have the same electron configuration $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $3d^8$ $4s^2$.

C The $^{64}_{30}Zn^{2+}$ ion has fewer neutrons in its nucleus than the $^{66}_{30}Zn^{2+}$ ion.

D The $^{66}_{30}Zn^{2+}$ ion will be deflected more than the $^{64}_{30}Zn^{2+}$ ion in an electric field of the same strength.

6 Gaseous particle **X** has an atomic number *n* and a charge of +2.
Gaseous particle **Y** has an atomic number of (*n* + 1) and is isoelectronic with **X**.

Which of the following statements correctly describes **X** and **Y**?

A **X** has a larger radius than **Y**.

B **X** has a larger nuclear charge than **Y**.

C **X** has a charge of +2 and **Y** has a charge of +1.

D **X** requires more energy than **Y** when a further electron is removed from each particle.

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7 In microwave ovens, the wave energy produced is absorbed by certain polar molecules.

Which of the following would absorb microwave energy?

A cyclohexane

B cyclohexene

C trichloromethane

D tetrachloromethane

8 For the pairs of species shown below, in which does the first species have a larger bond angle than the second?

A CH_2Cl_2, OCl_2

B PH_3, NH_3

C SO_3^{2-}, CO_3^{2-}

D BrF_2^{-}, $BeCl_2$

9 In which sequences are the molecules quoted in order of decreasing boiling points?

A $CH_3(CH_2)_3CH_3$, $(CH_3)_2CHCH_2CH_3$, $CH_3C(CH_3)_2CH_3$

B $AlBr_3$, $AlCl_3$, AlF_3

C SO_2, SiO_2, CO_2

D H_2O, NH_3, HF

10 Which of the following has an exothermic enthalpy change?

A $Ca\,(g) \rightarrow Ca^{2+}\,(g) + 2e^-$

B $Ca\,(s) + \frac{1}{2}O_2\,(g) \rightarrow CaO\,(s)$

C $CaO\,(s) \rightarrow Ca^{2+}\,(g) + O^{2-}\,(g)$

D $\frac{1}{2}O_2\,(g) \rightarrow O\,(g)$

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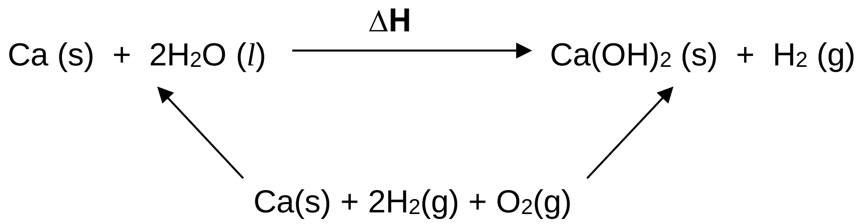
11 The standard enthalpy change of reaction between calcium and water, Δ**H**, can be determined with the help of the energy cycle below:



The standard enthalpy change of formation of $Ca(OH)_2(s)$ is −986 kJ mol^{-1} while the standard enthalpy change of combustion of $H_2(g)$ is −286 kJ mol^{-1}.

Which of the following is the correct value for Δ**H**?

A +700 kJ mol^{-1}

B –700 kJ mol^{-1}

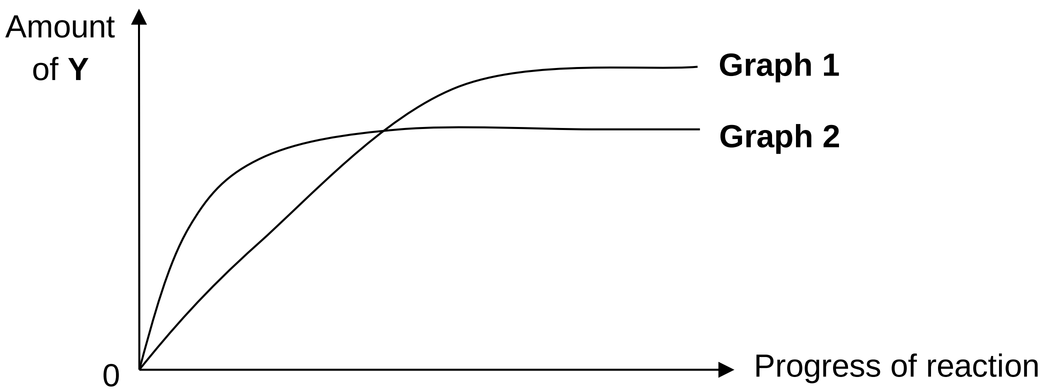
C +414 kJ mol^{-1}

D –414 kJ mol^{-1}

12 The graph for the equilibrium,

$$\mathbf{W}\,(g) + \mathbf{X}\,(g) \rightleftharpoons 3\mathbf{Y}\,(g) \qquad \Delta H < 0$$

is given below:



Which of the following changes could **not** account for the change from **Graph 1** to **Graph 2**?

A Increase in pressure

B Increase in temperature

C Decrease in amount of **X**

D Decrease in volume of vessel

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13 Consider the following equilibrium system:

$$H_2\,(g) + I_2\,(g) \rightleftharpoons 2HI\,(g) \qquad \Delta H = +53 \text{ kJ mol}^{-1}$$

Which of the following change is **not** correct?

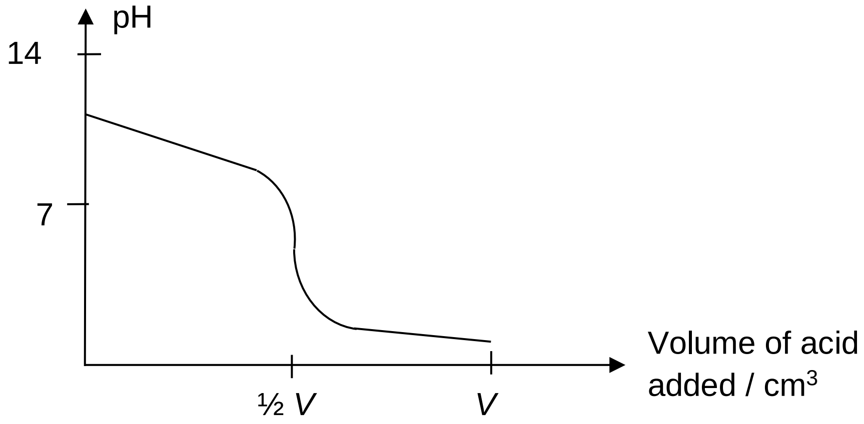
A Increasing the pressure will increase the rate of the backward reaction.

B Increasing the mass of H_2 will cause the equilibrium constant to increase.

C Increasing temperature increases the rate constant and equilibrium constant.

D Rate of forward reaction is equal to rate of backward reaction when equilibrium is reached.

14 The graph shows the change in pH when 0.25 mol dm^{-3} acid is gradually added to V cm^3 of 0.25 mol dm^{-3} base.



Which pair of solutions will give the result as shown in the graph?

A HNO_3 and NH_3

B H_2SO_4 and NH_3

C CH_3COOH and $Ca(OH)_2$

D $CH_2(COOH)_2$ and NaOH

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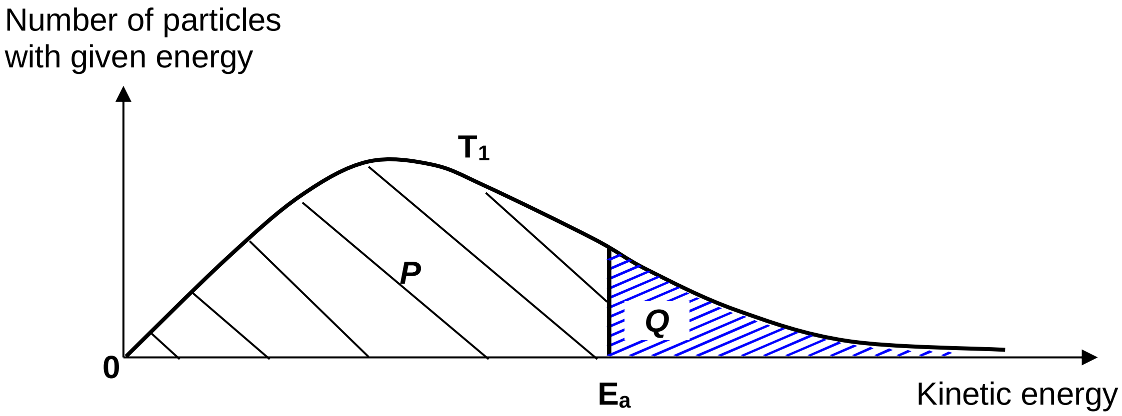
15 The diagram shows the Boltzmann distribution of the speeds of the molecules of a gas at temperature, $\mathbf{T_1}$. The letters ***P*** and ***Q*** refer to the separate and differently shaded areas.



If the above gas is cooled to a new temperature, $\mathbf{T_2}$, how would the differently shaded areas, ***P*** and ***Q*** change in relation to total area of the curve, ***P*** + ***Q***?

	P	***Q***	***P*** + ***Q***
A	Increase	Decrease	Unchanged
B	Decrease	Increase	Unchanged
C	Increase	Increase	Increase
D	Decrease	Decrease	Decrease

16 Which factor decreases from Na to Si (excluding Mg and A*l*) and also from Si to P?

A Ionic radius

B Melting point

C Electrical conductivity

D Reactivity with oxygen

17 Element **G** is in the third period of the Periodic Table. The chloride of **G** has a simple molecular structure while the oxide of **G** has a giant ionic structure.

Which of the following statements is true about **G**?

A The atomic radius of **G** is smaller than that of chlorine.

B The first ionisation energy of **G** is higher than that of Mg.

C The chloride of **G** dissolves in water to give a neutral solution.

D The oxide of **G** reacts with aqueous sodium hydroxide and aqueous hydrochloric acid.

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18 **J**, **K** and **L** are elements in Period 3. **K** has a larger ionic radius than **J**, and **L** has a less endothermic first ionisation energy than **K**.

What are elements **J**, **K** and **L**?

	J	**K**	**L**
A	P	S	C*l*
B	P	A*l*	Mg
C	A*l*	Mg	Si
D	A*l*	P	S

19 One of the isomers of dimethylbenzoic acid is 2,3-dimethylbenzoic acid.

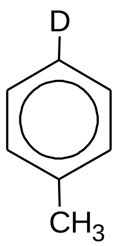
How many different isomers are possible for dimethylbenzoic acid, inclusive of 2,3-dimethylbenzoic acid?

A 5 **B** 6 **C** 7 **D** 8

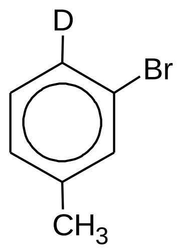
20 Deuterium, D, is a heavy isotope of hydrogen. 4-methyldeuteriobenzene is reacted with bromine and a suitable catalyst so that only monobromination takes place.

Assume that the carbon–deuterium bond is broken as easily as a carbon–hydrogen bond.

What proportion of the product will be 2-bromo-4-methyldeuteriobenzene?



4-methyldeuteriobenzene



2-bromo-4-methyldeuteriobenzene

A 0% **B** 33% **C** 40% **D** 67%

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21 Ozone depletion potential, ODP, is a measure of the extent of degradation to the ozone layer that an ozone-depleting substance can cause per unit mass of the substance.

The following table shows the ODP of three ozone-depleting substances.

substances	C_2F_5Cl	$C_2F_4Cl_2$	$C_2F_4Br_2$
ODP	0.6	1.0	6.0

Based on the above information, which of the following statements is correct regarding ozone-depleting substances?

A $C_2F_4Br_2$ has a higher ODP than $C_2F_4Cl_2$ because $C_2F_4Br_2$ has a higher molar mass.

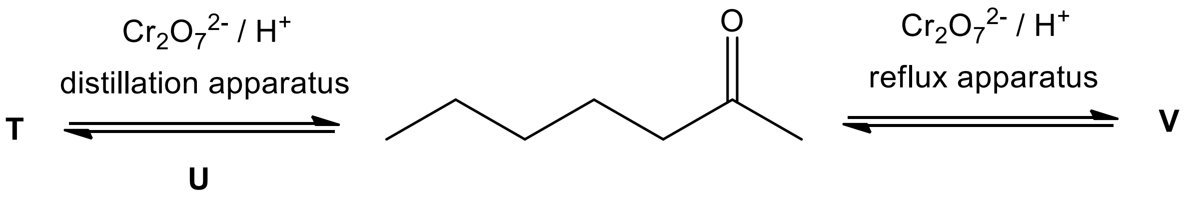
B $C_2F_4Br_2$ has a higher ODP than $C_2F_4Cl_2$ because $C_2F_4Br_2$ has a higher percentage composition by mass of halogen.

C $C_2F_4Br_2$ has a higher ODP than $C_2F_4Cl_2$ because C–Br has lower bond energy than C–C*l*.

D Fluorine radical is more harmful to the ozone layer than chlorine radicals or bromine radicals.

22 One of the chemicals giving blue cheese its unique aroma is heptan-2-one.

The diagram shows reactions involving heptan-2-one.



Which is the correct identification of compound **T**, reagent **U** and compound **V**?

	compound **T**	reagent **U**	compound **V**
A	heptane	$NaBH_4$	heptanoic acid
B	heptan-2-ol	$NaBH_4$	heptanoic acid
C	heptanal	hydrogen gas	heptan-2-one
D	heptan-2-ol	$LiAlH_4$ in dry ether	heptan-2-one

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23 In which of the following reactions would each molecule of the organic product have the same number of oxygen atoms as each molecule of the organic reactant?

A reacting 1-chloropropane with aqueous potassium hydroxide

B reacting 2-chloropropane with ethanolic potassium hydroxide

C reacting propan-1-ol with acidified potassium dichromate(VI)

D reacting 2-phenylpropan-1-ol with acidified potassium manganate(VII)

24 The compound $C_4H_6O_2$ gives butter its distinctive flavour.

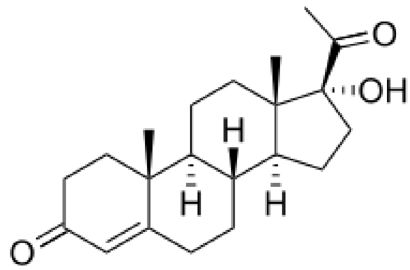
It reacts with hydrogen cyanide to form $C_6H_8N_2O_2$ but does not form a silver mirror with ammoniacal silver nitrate.

What is the structural formula of this compound in butter?

A CH_3COCH_2CHO **B** $CH_3COCOCH_3$

C $CH_3COCH{=}CHOH$ **D** $CH_2{=}CHCOCH_2OH$

25 Steroidogenesis is the biological pathway by which cholesterol is transformed into other biological steroids. One of the intermediates produced during steroidogenesis in human body is 17α-hydroxyprogesterone.



Which of the following statement is true for one molecule of 17α-hydroxyprogesterone?

A It turned aqueous potassium dichromate(VI) green.

B It produced one molecule of hydrogen gas when reacted with sodium metal.

C It can undergo oxidation, reduction, addition, substitution and elimination reactions.

D It produced an organic product with two more hydrogen atoms when reacted with hydrogen gas.

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Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

26 Beryllium difluoride, BeF_2, reacts readily with trimethylamine, $(CH_3)_3N$, to form a stable addition product. Nitrogen trifluoride has no reaction with trimethylamine.

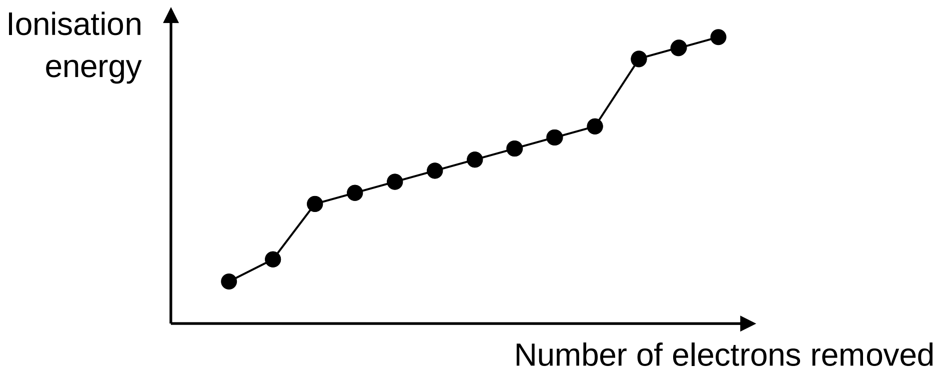
Which of the following statements are true?

1 Nitrogen trifluoride does not react as the nitrogen atom lacks energetically accessible orbitals for reaction.

2 The bond angle in the addition product is 109.5^{o}.

3 The molar ratio for the reaction between BeF_2 and $(CH_3)_3N$ is 1:1.

27 The graph shows the first thirteen ionisation energies for element **Z**.



What can be deduced about element **Z** from the graph?

1 The oxide of **Z** dissolves in water to form an alkaline solution.

2 The chloride of **Z** undergoes hydrolysis to form a solution of pH 3.

3 The element **Z** is an element in Period 2.

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The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

28 Which of the following reactions are hydrolysis reactions?

1 conversion of $CH_3CO_2CH_2CH_3$ into CH_3CO_2H and CH_3CH_2OH

2 conversion of CH_3CH_2CN into $CH_3CH_2CO_2^-Na^+$

3 conversion of P_4O_{10} into H_3PO_4

29 Alkanes can be prepared from chloroalkanes by heating under reflux with sodium (in ether) according to the equation:

$$2RCl + 2Na \rightarrow R\text{–}R + 2NaCl$$

Which alkanes will be produced if a mixture containing equal amounts of CH_3CH_2Cl and $CH_3CHClCH_3$ are used?

1 $CH_3CH_2CH_2CH_3$

2 $CH_3CH_2CH(CH_3)_2$

3 $CH_3CH(CH_3)CH(CH_3)_2$

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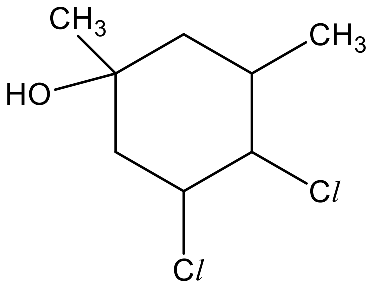
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

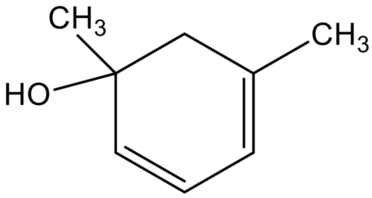
No other combination of statements is used as a correct response.

30 The following compound was heated with ethanolic sodium hydroxide.



Which of the following could represent the structure of the organic product?

1



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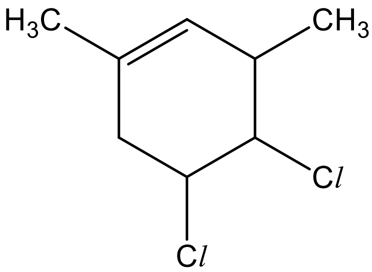

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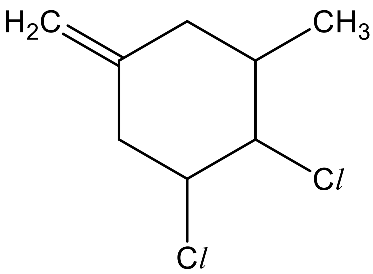

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ANSWERS

1	D	11	D	21	B
2	A	12	C	22	D
3	C	13	B	23	B
4	C	14	B	24	B
5	C	15	A	25	C
6	A	16	C	26	B
7	C	17	D	27	D
8	A	18	D	28	A
9	A	19	B	29	A
10	B	20	A	30	D

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